

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1517

Chico A 138 kV
August 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1517. The Interconnection Customer proposes a 95.5 MW wind facility interconnecting at the Chico A 138 kV substation in the Virginia area. The request was received on July 06, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 544 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,905,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Acec Lewiston Substation-Indian Flat 138 kV	York Canyon-Chico A 138 kV	131.7	209	19.0
York Canyon-Chio A 138 kV L	Mira Vista-Yarnen 345 kV	118.4	417	40.7
Dunnsville-Northern Neck 138 kV	York Canyon Substation-Chico A 138 kV	110.7	215	24.5
Sutter-Dunnsville 138 kV L	ChicoA-York Canyon Substation 138 kV	109.8	209	24.5
Indian Flat-Sutter 138 kV	Yor Canyon-Chico A 138 kV	107.4	206	24.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Acec Lewiston Substation-Indian Flat 138 kV	Rebuild line with higher-capacity conductor	\$8,979,000	53% (\$4,770,000)
York Canyon-Chio A 138 kV L	Rebuild line with higher-capacity conductor	\$11,829,000	22% (\$2,637,000)
Dunnsville-Northern Neck 138 kV	Rebuild line with higher-capacity conductor	\$7,061,000	100% (\$7,061,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sutter-Dunnville 138 kV	Reconductor line	\$1,909,000	100% (\$1,909,000)
Indian Flat-Sutter 138 kV Line	Reconductor line	\$2,528,000	100% (\$2,528,000)
	Total allocated to project		\$18,905,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.